Křivočaré souřadnice

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Křivočaré souřadnice

M … prostor

$\mathcal{F}M$ … funkce na M $\qquad \mathcal{F}M \ni f : M \to \mathbb{R}$

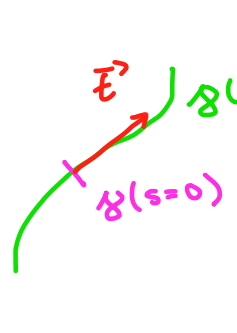
$\mathcal{T}M$ … vektorové pole na M $\qquad \mathcal{T}M \ni \vec{t} : \vec{t}|_m \in V$

$\mathcal{T}^*M$ … kovektorové pole na M $\qquad \mathcal{T}^*M \ni \underset{\sim}{\alpha} : \underset{\sim}{\alpha}|_m \in V^*$

$\mathcal{T}^k_\ell M$ … tenzorové pole na M $\qquad \mathcal{T}^k_\ell M \ni T : M \to T^k_\ell$

tenzor typu (k,ℓ)

M … souřadnice x^i $\quad i \in \{1, \dots, \dim M\}$



$s \in \mathbb{R}$

$\vec{t} = \left.\frac{d\gamma}{ds}\right|_{s=0}$

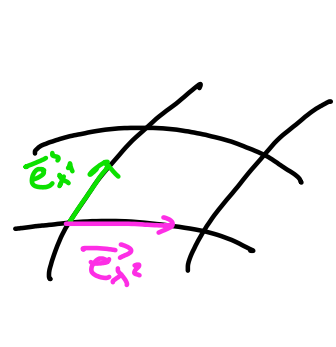
$f \in \mathcal{F}M \qquad D_{\vec{t}} f = \frac{df(\gamma(s))}{ds} = (\vec{t} \cdot \nabla) f$

směrová derivace

skalárový derivace f podle vektoru

nezávisí na γ, ale pouze na tečně

$\Rightarrow \quad D_{\vec{t}} \longleftrightarrow \vec{t}$ je obousměrný!



$D_{\vec{e}_{x^1}} f = \frac{df(x^1+s, x^2, \dots)}{ds} = \frac{\partial f}{\partial x^1}$

$\Rightarrow \quad D_{\vec{e}_{x^j}} = \frac{\partial}{\partial x^j}$

$\Rightarrow \quad \left\{\frac{\partial}{\partial x^j}\right\}_{i=1}^{\dim M}$ … báze $\mathcal{T}M$

$\underset{\sim}{\alpha} \in \mathcal{T}^*M, \ \vec{t} \in \mathcal{T}M \qquad \underset{\sim}{\alpha}(\vec{t}) \in \mathbb{R}$

$df \in \mathcal{T}M \qquad df(\vec{t}) = \frac{\partial f}{\partial x^i} t^i$

$x^i : M \to \mathbb{R} \ \Rightarrow \ dx^i \ \Rightarrow \ \{dx^i\}_{i=1}^{\dim M}$

$dx^i\left(\frac{\partial}{\partial x^j}\right) = \frac{\partial x^i}{\partial x^k}\left(\frac{\partial}{\partial x^j}\right)^k = \delta^i_j$

$\Rightarrow$ báze jsou duální

Transformace

$x^i \to x'^j$

$\frac{\partial}{\partial x'^j} = \frac{\partial x^i}{\partial x'^j}\frac{\partial}{\partial x^i} \qquad dx'^j = \frac{\partial x'^j}{\partial x^i} dx^i$

$T(\vec{t}, \underset{\sim}{\alpha}) = T\left(t^i \frac{\partial}{\partial x^i}, \alpha_j dx^j\right) = t^i \alpha_j \, T\left(\frac{\partial}{\partial x^i}, dx^j\right)$

$= t^i \alpha_j T^j_i$

$\Rightarrow \quad T'^k_\ell = T\left(\frac{\partial}{\partial x'^\ell}, dx'^k\right) = \frac{\partial x'^k}{\partial x^i}\frac{\partial x^j}{\partial x'^\ell} T^i_j$

Př

$\vec{t} \cdot dx^i = dx^i \cdot \vec{t} = \vec{t}(x^i) = t^i$

Metrika

$g = g_{ij} dx^i dx^j$

- tenzorové pole typu $(0,2)$ – 2-krát kovariantní tenzor je v každém bodě nedegenerovaný, pozitivně definitní, symetrický

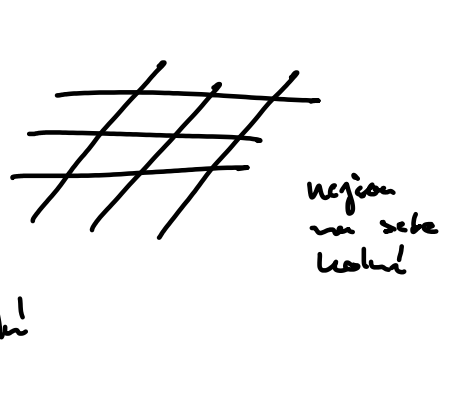
1) $\mathbb{E}^3$ + kartézské souřadnice $\quad g_{ij} = \delta_{ij}$

$\Rightarrow$ vektory ortonormální $\quad \delta_{ij} = \frac{\partial x^k}{\partial x'^i}\frac{\partial x^\ell}{\partial x'^j}\delta_{k\ell}$

$\Rightarrow \quad A^T A = \mathbb{1}$

2) $\mathbb{E}^3$ + lin. souřadnice

$g_{ij} = \text{konst}$



$\Rightarrow$ nemusí mít diagonálu!

3) $\mathbb{E}^3$ + ortogonální souřadnice

$g_{ij} = \begin{pmatrix} h_1^2 & & 0 \\ & h_2^2 & \\ 0 & & h_3^2 \end{pmatrix}$ $\qquad k_i$ … Lamého koeficienty

$k_i^2 = g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^i}\right)$

4) ortonormální souřadnice – trivialita

ortogonální $\to \quad \vec{e}_i = \frac{1}{h_i}\frac{\partial}{\partial x^i}$ (nesčítá se!)

$g(\vec{e}_i, \vec{e}_j) = \delta_{ij}$

Konstrukce souřadnic

Dim = 2

$\underset{\mathbb{E}^2}{(x,y)} \to z = x + iy \to f(z) = u(x,y) + i\,v(x,y)$

Předpoklady:

1) $f'(z) = \frac{df}{dz}(z) \neq 0$

2) f je holomorfní $\to$ konformní zobrazení $\to$ zachovává úhly

Cauchy Riemann $\quad u_{,x} = v_{,y}$, $\quad u_{,y} = -v_{,x}$

$df \Leftrightarrow \nabla f$

$g^{-1}(du, dv) = g^{-1}(u_{,x}dx + u_{,y}dy, \ v_{,x}dx + v_{,y}dy)$

kanonická metrika = inverze

$= u_{,x}v_{,x} + u_{,y}v_{,y}$

$= -u_{,y}u_{,x} + u_{,y}u_{,x} = 0$

opačný přístup – cuklidovská geometrie funguje v u a v

$\frac{\partial}{\partial x} = \frac{\partial u}{\partial x}\frac{\partial}{\partial u} + \frac{\partial v}{\partial x}\frac{\partial}{\partial v}$

$\frac{\partial}{\partial y} = \frac{\partial u}{\partial y}\frac{\partial}{\partial u} + \frac{\partial v}{\partial y}\frac{\partial}{\partial v}$

$g\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) = \frac{\partial u}{\partial x}\cdot\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\cdot\frac{\partial v}{\partial y} = 0$

Př

$z \to e^z$

$x + iy \to e^x e^{iy} = e^x(\cos y + i \sin y)$

$u = e^x \cos y \qquad v = e^x \sin y$

$\Rightarrow$ odpovídá transformaci do tak trochu polárních souřadnic

z $x = \text{konst.}$ jsou kružnice

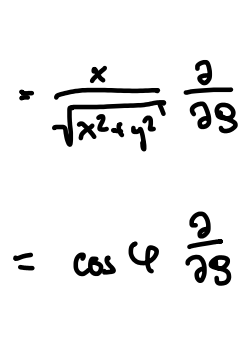
z $y = \text{konst.}$ jsou polopřímky z počátku

Kartézské → polární

$\mathbb{E}^3$ 2d souřadnice $\to$ stahování (polární $\to$ cylindrické)

$\qquad\qquad\qquad\ \ \to$ rotace (polární $\to$ sférické)

2d eliptické souřadnice



$\to$ protáhlé elipsoidální souřadnice

$\to$ zploštělé elipsoidální souřadnice

Obecnější v $\mathbb{E}^3$

- trojosý elipsoidální souřadný systém

Cylindrické souřadnice

$x = \varrho \cos\varphi \qquad \varrho = \sqrt{x^2+y^2}$

$y = \varrho \sin\varphi \qquad \varphi = \text{arctg}\,\frac{y}{x}$

$z = z$

$dx = \cos\varphi \, d\varrho - \varrho \sin\varphi \, d\varphi$

$dy = \dots$

$dz = dz$

$\frac{\partial}{\partial x} = \frac{\partial \varrho}{\partial x}\frac{\partial}{\partial \varrho} + \frac{\partial \varphi}{\partial x}\frac{\partial}{\partial \varphi}$

$= \frac{x}{\sqrt{x^2+y^2}}\frac{\partial}{\partial \varrho} - \frac{y}{x^2+y^2}\frac{\partial}{\partial \varphi}$ } v hybridní fázi

$= \cos\varphi \frac{\partial}{\partial \varrho} - \frac{\sin\varphi}{\varrho}\frac{\partial}{\partial \varphi}$

$g = dxdx + dydy + dzdz$

$= (\cos\varphi \, d\varrho - \varrho \sin\varphi \, d\varphi)(\cos\varphi \, d\varrho - \varrho \sin\varphi \, d\varphi)$

$+ (\sin\varphi \, d\varrho + \varrho \cos\varphi \, d\varphi)(\sin\varphi \, d\varrho + \varrho \cos\varphi \, d\varphi)$

$+ dzdz$

$= d\varrho d\varrho + \varrho^2 d\varphi d\varphi + dzdz$

$- \varrho \sin\varphi\cos\varphi\,(d\varphi \cancel{d\varrho} + d\varrho \cancel{d\varphi} - d\varphi \cancel{d\varrho} - d\varrho \cancel{d\varphi})$

$= \underline{1} d\varrho d\varrho + \underline{\varrho^2} d\varphi d\varphi + \underline{1} dzdz$

Lamého koeficienty

$k_\varrho = 1 \qquad k_\varphi = \varrho \qquad k_z = 1$

Sférické souřadnice

- analogicky:

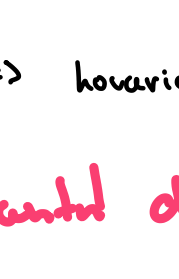
${}^{3D}g = drdr + r^2 d\theta d\theta + r^2 \sin^2\theta \, d\varphi d\varphi$

$r = 2 \to 2D = \text{sféra}$

$\Rightarrow dr = 0$

$\Rightarrow \ {}^{2D}g = d\theta d\theta + \sin^2\theta \, d\varphi d\varphi$

$\to$ porovnej s polárními souřadnicemi



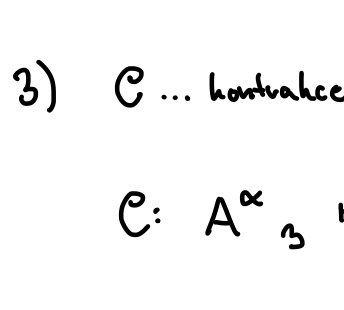
$\to$ po přeměně měří $\to$ jinak

Derivování tenzorů

$T = T^{ab}_{cd} \underbrace{\frac{\partial}{\partial x^a}\frac{\partial}{\partial x^b} dx^c dx^d}_{\text{obecně nejsou homogenní pole}}$

$\Rightarrow$ při derivování se tady musí zderivovat

vnímá:



$\left.\frac{dT(\gamma(s))}{ds}\right|_{s=0} = \vec{t} \cdot \nabla T$

$\lim_{\delta \to 0} \frac{T(\gamma(s-\delta)) - T(\gamma(s))}{\delta}$ } ale odečítat tenzory ve dvou bodech je nečisté!

$\Rightarrow$ vede na zeesti!

$\to$ musíme na to jinak

derivací tenzoru by měl být opět tenzor

$\Rightarrow$ kovariantní derivace axiomaticky

Kovariantní derivace axiomaticky

1) $\nabla(A + B) = \nabla A + \nabla B$

A, B jsou tenzory stejného typu

2) $\nabla(AB) = (\nabla A)B + A(\nabla B)$ ← hledíme pořadí obou tenzorů

speciálně

$\nabla(fA) = (\nabla f)A + f(\nabla A)$

3) $\mathcal{C}$ … kontrakce

$\mathcal{C} : A^\alpha{}_\beta \mapsto A^\alpha{}_\alpha$

$\nabla(\mathcal{C}A) = \mathcal{C}\,\nabla A$

4) ∇f … gradient

Takovýchto operací je nekonečně mnoho

Př

$\mathbb{E}^d$ … lineární souřadnice

$\nabla dx^{\bar{m}} = 0 \quad \Leftrightarrow \quad \nabla \frac{\partial}{\partial x^{\bar{m}}} = 0$ ← lineární souřadnice

$\nabla T = \nabla\left(T^{\bar{a}\bar{b}\dots}_{\bar{c}\bar{d}\dots}\frac{\partial}{\partial x^{\bar{a}}}\frac{\partial}{\partial x^{\bar{b}}}\cdots dx^{\bar{c}} dx^{\bar{d}}\dots\right)$

$= \left(\nabla T^{\bar{a}\bar{b}}_{\bar{c}\bar{d}}\right)\frac{\partial}{\partial x^{\bar{a}}}\frac{\partial}{\partial x^{\bar{b}}}\cdots dx^{\bar{c}} dx^{\bar{d}}\dots$

$= T^{\bar{a}\bar{b}}_{\bar{c}\bar{d}\dots,\bar{k}}\, dx^{\bar{k}} \frac{\partial}{\partial x^{\bar{a}}}\frac{\partial}{\partial x^{\bar{b}}}\cdots dx^{\bar{c}} dx^{\bar{d}}\dots$

Pozn.

$a^j_{\ i} dx^i \otimes \frac{\partial}{\partial x^j}$ ← složky tenzoru

$dx^i \cdot \frac{\partial}{\partial x^j} = \frac{\partial x^i}{\partial x^j} = \delta^i_j$

$\vec{a} \cdot \underset{\sim}{\alpha} = \left(a^j \frac{\partial}{\partial x^j}\right)\cdot(\alpha_i dx^i) = a^j \alpha_i \frac{\partial}{\partial x^j}\cdot dx^i = a^j \alpha_j$

Kovariantní derivace

$\nabla \vec{a} = a^i{}_{;j}\, dx^j \frac{\partial}{\partial x^i} = (\nabla\vec{a})^i{}_j\, dx^j \frac{\partial}{\partial x^i}$ } značení

$= \nabla_j a^i dx^j \frac{\partial}{\partial x^i}$

Parametrizace

$\nabla \vec{a} = \nabla\left(a^i \frac{\partial}{\partial x^i}\right) = (\nabla a^i)\frac{\partial}{\partial x^i} + a^i\left(\nabla \frac{\partial}{\partial x^i}\right)$

tady musí být typu $(1,1)$, aby to bylo tenzorový typ

$= a^i{}_{,j}\, dx^j \frac{\partial}{\partial x^i} + a^i \Gamma^k_{ji}\, dx^j \frac{\partial}{\partial x^k}$

$= \underbrace{\left(a^i{}_{,j} + a^\ell \Gamma^i_{j\ell}\right)}_{=:\ a^i{}_{;j}} dx^j \frac{\partial}{\partial x^i}$

$\nabla\underset{\sim}{\alpha} = \nabla(\alpha_j dx^j) = \alpha_{j,i}\, dx^i dx^j - \alpha_k \overset{\text{záleží na pořadí!}}{\tilde{\Gamma}^k_{ij}} dx^i dx^j$

$= \underbrace{\left(\alpha_{j,i} - \alpha_k \tilde{\Gamma}^k_{ij}\right)}_{=:\ \alpha_{j;i}} dx^i dx^j$

$0 = \nabla\left(dx^i \cdot \frac{\partial}{\partial x^j}\right) = -\tilde{\Gamma}^i_{k\ell}\, dx^k \underbrace{dx^\ell \cdot \frac{\partial}{\partial x^j}}_{\delta^\ell_j}$

$+ \underbrace{dx^i \cdot \Gamma^\ell_{kj}}_{\delta^i_\ell}\, dx^k \frac{\partial}{\partial x^\ell}$

$= -\tilde{\Gamma}^i_{kj}\, dx^k + \Gamma^i_{kj}\, dx^k$

$\Rightarrow \quad \tilde{\Gamma} = \Gamma \qquad$ Christoffelovy symboly

$T^{ab}_{cd;j} = T^{ab}_{cd,j} + \Gamma^a_{jk} T^{kb}_{cd} + \Gamma^b_{jk} T^{ak}_{cd}$

$- \Gamma^k_{jc} T^{ab}_{kd} - \Gamma^k_{jd} T^{ab}_{ck}$

$\frac{\partial}{\partial x^j}\cdot\left(\nabla \frac{\partial}{\partial x^i}\right)\cdot dx^k = \Gamma^k_{ji}$

$\frac{\partial}{\partial x^i}\cdot(\nabla dx^j)\cdot\frac{\partial}{\partial x^k} = \Gamma^j_{ki}$

Př $\quad \mathbb{E}^d$ a lineární souřadnice $x^{\bar{i}}$

$\nabla dx^{\bar{i}} = 0 = 0\,\nabla \frac{\partial}{\partial x^{\bar{i}}}$

$\Gamma^k_{ji} = \frac{\partial}{\partial x^j}\cdot\left(\nabla \frac{\partial x^{\bar{m}}}{\partial x^i}\right)\frac{\partial}{\partial x^{\bar{m}}}\cdot dx^k$

$= \frac{\partial}{\partial x^j}\cdot\frac{\partial^2 x^{\bar{m}}}{\partial x^n \partial x^i}\frac{\partial}{\partial x^m} dx^n \frac{\partial}{\partial x^{\bar{m}}}\cdot dx^k$

$= \frac{\partial^2 x^{\bar{m}}}{\partial x^j \partial x^i}\ \frac{\partial x^k}{\partial x^{\bar{m}}}$

$\Gamma^j_{ki} = \frac{\partial}{\partial x^k}(\nabla dx^j)\frac{\partial}{\partial x^i} = \frac{\partial}{\partial x^k}\left(\nabla \frac{\partial x^j}{\partial x^{\bar{m}}} dx^{\bar{m}}\right)\frac{\partial}{\partial x^i}$

$= \frac{\partial^2 x^j}{\partial x^k \partial x^{\bar{m}}}\ \frac{\partial x^m}{\partial x^i} = \frac{\partial^2 x^i}{\partial x^{\bar{\ell}} \partial x^{\bar{m}}}\ \frac{\partial x^{\bar{\ell}}}{\partial x^k}\ \frac{\partial x^{\bar{m}}}{\partial x^i}$

Obecný případ

1) Symetrie $\quad \Gamma^i_{jk} = \Gamma^i_{kj} \quad \equiv$ beztorzovost

2) kompatibilita s metrikou $\quad \nabla g = 0$

$((\nabla g)_{abc}\, dx^c dx^a dx^b)$

$0 = g_{ab;c} = g_{ab,c} - \cancel{\Gamma^d_{ca} g_{db}} - \cancel{\Gamma^d_{cb} g_{ad}}$ ⊖

$0 = g_{bc;a} = g_{bc,a} - \Gamma^d_{ab} g_{dc} - \cancel{\Gamma^d_{ac} g_{bd}}$ ⊕

$0 = g_{ca;b} = g_{ca,b} - \cancel{\Gamma^d_{bc} g_{da}} - \Gamma^d_{ba} g_{cd}$ ⊕

$\Gamma^d_{ab} g_{dc} = g_{bc,a} + g_{ca,b} - g_{ab,c}$

$\Gamma^f_{ab} = \frac{1}{2} g^{cf}(g_{bc,a} + g_{ca,b} - g_{ab,c})$

Speciálně v $\mathbb{E}^d$ s ortogonálními souř.

$g = \begin{pmatrix} h_1^2 & & 0 \\ & h_2^2 & \\ 0 & & \ddots \ h_d^2 \end{pmatrix}$

$k \neq m \neq n \neq k$

$\Gamma^k_{mn} = \frac{1}{2} g^{kk}(0 + 0 - 0) = 0$

$k \neq m$

$\Gamma^k_{mm} = \frac{1}{2} g^{kk}(0 + 0 - g_{mm,k})$

$= -\frac{1}{2}\frac{1}{h_k^2}(h_m^2)_{,k} = -\frac{h_m h_{m,k}}{h_k^2}$

$\Gamma^k_{km} = \frac{1}{2} g^{kk}(g_{km,k} + g_{kk,m} - g_{km,k})$

$= \frac{h_k h_{k,m}}{h_k^2}$

$\sum_k \Gamma^k_{km} = \Big[\ln\underbrace{(h_1 \cdot h_2 \cdot h_3 \cdots)}_{\sqrt{\det g}}\Big]_{,m}$

Pozn. $\quad$ ON báze na $\mathbb{E}^d$

$e_{\hat{i}} = \frac{1}{h_i}\frac{\partial}{\partial x^i}$ (nesčítá se)

$e^{\hat{j}} = h_j\, dx^j$

$\vec{a} = a^i \frac{\partial}{\partial x^i} = a^{\hat{i}} e_{\hat{i}}$

$h_i a^i = a^{\hat{i}} = a_{\hat{i}} = \frac{1}{h_i} a_i$

Operátory v orto(n)gonální souř.

$(\nabla f)_m = \nabla_m f = f_{,m} \ \Rightarrow \ (\nabla f)_{\hat{m}} = \frac{f_{,m}}{h_m} = (\nabla f)^{\hat{m}}$

$\nabla\cdot\vec{a} = \nabla_i a^i = a^i{}_{;i} = a^i{}_{,i} + \underbrace{\Gamma^i_{ij}}_{\text{sčítá se}} a^j$

$= a^j{}_{,j} + \frac{(h_1 \cdots h_d)_{,j}}{h_1 \cdots h_d} a^j$

$= \frac{(a^j h_1 \cdots h_d)_{,j}}{h_1 \cdots h_d}$

speciálně $d = 3$

$\Rightarrow \quad \nabla\cdot\vec{a} = \frac{1}{h_1 h_2 h_3}\left[(a^1 h_2 h_3)_{,1} + (a^2 h_1 h_3)_{,2} + (a^3 h_1 h_2)_{,3}\right]$

$= \frac{1}{h_1 h_2 h_3}\Big[\Big(\overset{\text{kovariantní}}{\frac{a_1}{h_1}} h_2 h_3\Big) + \dots\Big]$

$\Delta f = \nabla\cdot\nabla f = \underbrace{f^{;k}{}_{k} = f_{;k}{}^{k}}_{\text{můžu manipulovat s metrikou, jelikož } \nabla g = 0} =$

$= \frac{1}{h_1 h_2 h_3}\left[\left(\frac{h_2 h_3}{h_1} f_{,1}\right)_{,1} + \left(\frac{h_1 h_3}{h_2} f_{,2}\right) + \left(\frac{h_1 h_2}{h_3} f_{,3}\right)_{,3}\right]$

Vsuvka

Levi-Civitův symbol

$\varepsilon \in \mathcal{T}^0_d(\mathbb{E}^d)$

1) antisymetrický komplet

$\varepsilon_{i_1 \dots i_d} = \varepsilon_{[i_1 \dots i_d]}$

má plný rank

2) normalizace

$\varepsilon_{\hat{1} \dots \hat{d}} = 1 \qquad$ pro pravotočivě orientovanou bázi

$\varepsilon_{i_1 \dots i_d}\,\varepsilon^{i_1 \dots i_d} = d!$

$\Rightarrow$ co platí pro $\varepsilon_{1 \dots d}$?

$\varepsilon_{1 \dots d} = h_1 \cdot \dots \cdot h_d$

$\varepsilon^{1 \dots d} = \frac{1}{h_1} \cdot \dots \cdot \frac{1}{h_d}$

$\varepsilon_{1 \dots d}\, dx^1 \dots dx^d \ \to$ zajišťuje orientaci integračního elementu

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$(\nabla \times a)^k = \varepsilon^{kmn} a_{n;m} = \varepsilon^{kmn}\big(a_{n,m} - \underbrace{\Gamma^j_{nm} a_j}_{\to 0}\big)$

antisym. · sym.

$= \varepsilon^{kmn} a_{n,m}$

$(\nabla \times a)^1 = \varepsilon^{123}(a_{3,2} - a_{2,3}) = \frac{1}{h_1 h_2 h_3}(a_{3,2} - a_{2,3})$

$= \frac{1}{h_1 h_2 h_3}\left((h_3^2 a_3)_{,2} - (h_2^2 a_2)_{,3}\right)$

$(\nabla \times a)^{\hat{1}} = \frac{1}{h_2 h_3}\left((h_3 a_{\hat{3}})_{,2} - (h_2 a_{\hat{2}})_{,3}\right)$